240840116062

Practical 8 : Create interactive visualization using Plotly Express and Ploty Graph Objects

***Step 1 - import Required libraries ***

```
# import pandas
#import plotly Express
# import Plotly Graph Objects

import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
```

** Step 2: Create a sommon dataset **

```
# creating a dataset
data = {
    "Month" : ['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec'],
    "Product" : ['Laptop','Mobile','Tablet','Laptop','Mobile','Tablet','Laptop','Mobile','Tablet','Laptop','Mobile','Tablet'
    "Region" : ["North","South','East','West","North","South','East','West","North","South','East','West'],
    "Sales" : [120000, 135000, 150000, 145000, 160000, 175000, 180000, 170000, 190000, 210000, 230000, 250000],
    "Units_Sold" : [12, 30, 25, 14, 35, 30, 18, 38, 32, 21, 45, 40],
    "Profit" : [18000,27000,30000,22000,32000,35000,28000,34000,38000,33000,46000,50000]
}
df = pd.DataFrame(data)
df
```

	Month	Product	Region	Sales	Units_Sold	Profit
0	Jan	Laptop	North	120000	12	18000
1	Feb	Mobile	South	135000	30	27000
2	Mar	Tablet	East	150000	25	30000
3	Apr	Laptop	West	145000	14	22000
4	May	Mobile	North	160000	35	32000
5	Jun	Tablet	South	175000	30	35000
6	Jul	Laptop	East	180000	18	28000
7	Aug	Mobile	West	170000	38	34000
8	Sep	Tablet	North	190000	32	38000
9	Oct	Laptop	South	210000	21	33000
10	Nov	Mobile	East	230000	45	46000
11	Dec	Tablet	West	250000	40	50000

Next steps: Generate code with df | New interactive sheet

```
#Display first five rows .
df.head()
```

	Month	Product	Region	Sales	Units_Sold	Profit
0	Jan	Laptop	North	120000	12	18000
1	Feb	Mobile	South	135000	30	27000
2	Mar	Tablet	East	150000	25	30000
3	Apr	Laptop	West	145000	14	22000
4	May	Mobile	North	160000	35	32000

Next steps: Generate code with df | New interactive sheet

```
# dispaly the no of rows and column
df.shape
```

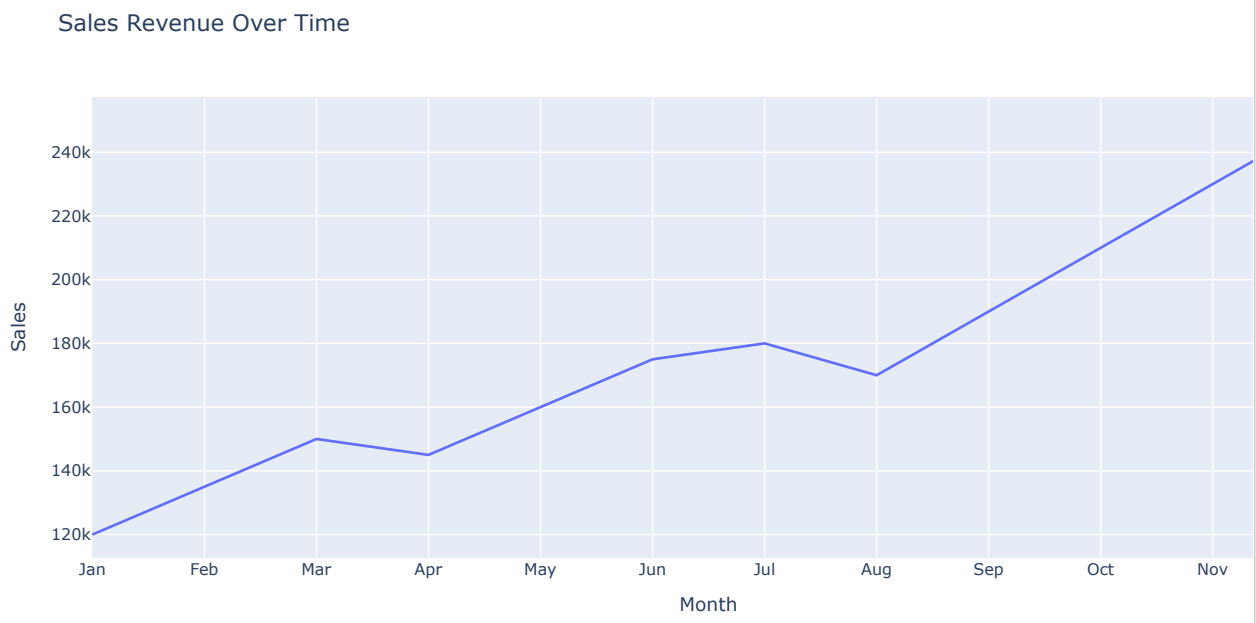
```
(12, 6)
```

```
# Display thr column names and datatypes
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   Month       12 non-null     object
 1   Product     12 non-null     object
 2   Region      12 non-null     object
 3   Sales       12 non-null     int64 
 4   Units_Sold  12 non-null     int64 
 5   Profit      12 non-null     int64 
dtypes: int64(3), object(3)
memory usage: 708.0+ bytes
```

Step 3 create an interactive line chart using ploty Express

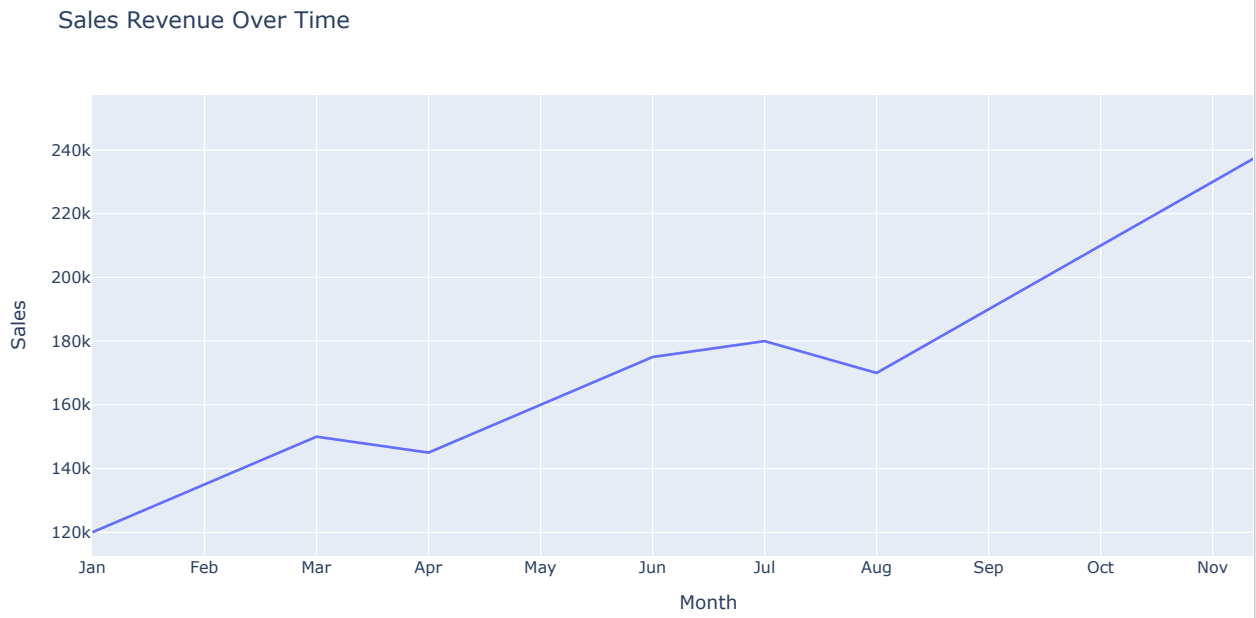
```
# How dose sales revenue change throughout the year?
fig = px.line(df, x="Month", y="Sales", title="Sales Revenue Over Time")
fig.show()
```



**step 4 : Improve the hover Information **

```
# create  a line chart using sales over Month.
# Add Additional columns to the hover information . Students can see products ,Region , Unit Sold and profit when they move

fig = px.line(df, x="Month", y="Sales", title="Sales Revenue Over Time",
              hover_name="Month", hover_data=["Units_Sold", "Profit"])
fig.show()
```



Step 5 : Create an interactive Bar chart

```
# Gourp the Dataset according to product.
df_grouped = df.groupby('Product').agg({'Sales': 'sum', 'Units_Sold': 'sum', 'Profit': 'sum'}).reset_index()
df_grouped
```

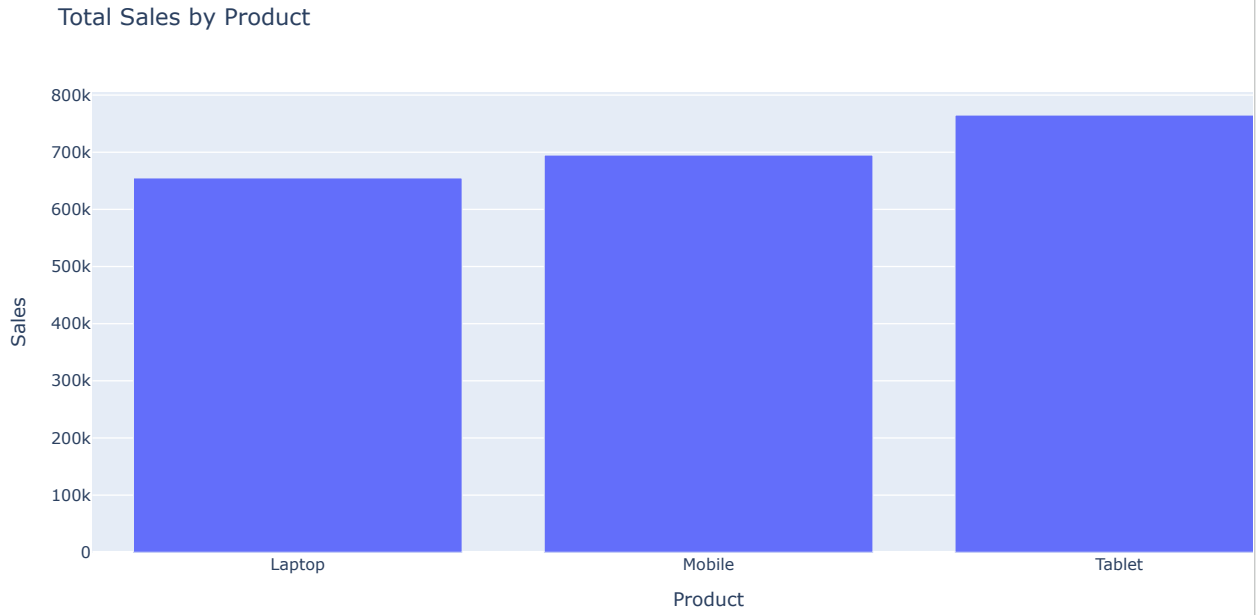
	Product	Sales	Units_Sold	Profit
0	Laptop	655000	65	101000
1	Mobile	695000	148	139000
2	Tablet	765000	127	153000

Next steps: Generate code with `df_grouped` | New interactive sheet

```
# calculate the total sales generated by each product
total_sales = df_grouped['Sales'].sum()
total_sales
```

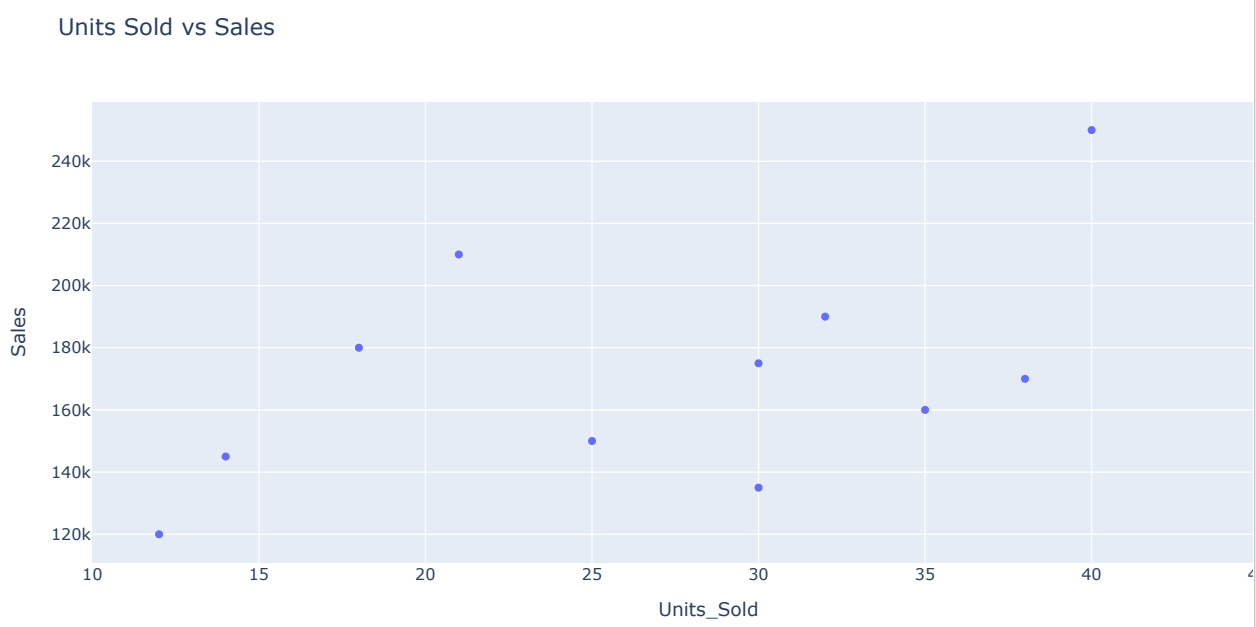
```
np.int64(2115000)
```

```
# Create an interractive bar chart
fig = px.bar(df_grouped, x='Product', y='Sales', title='Total Sales by Product')
fig.show()
```



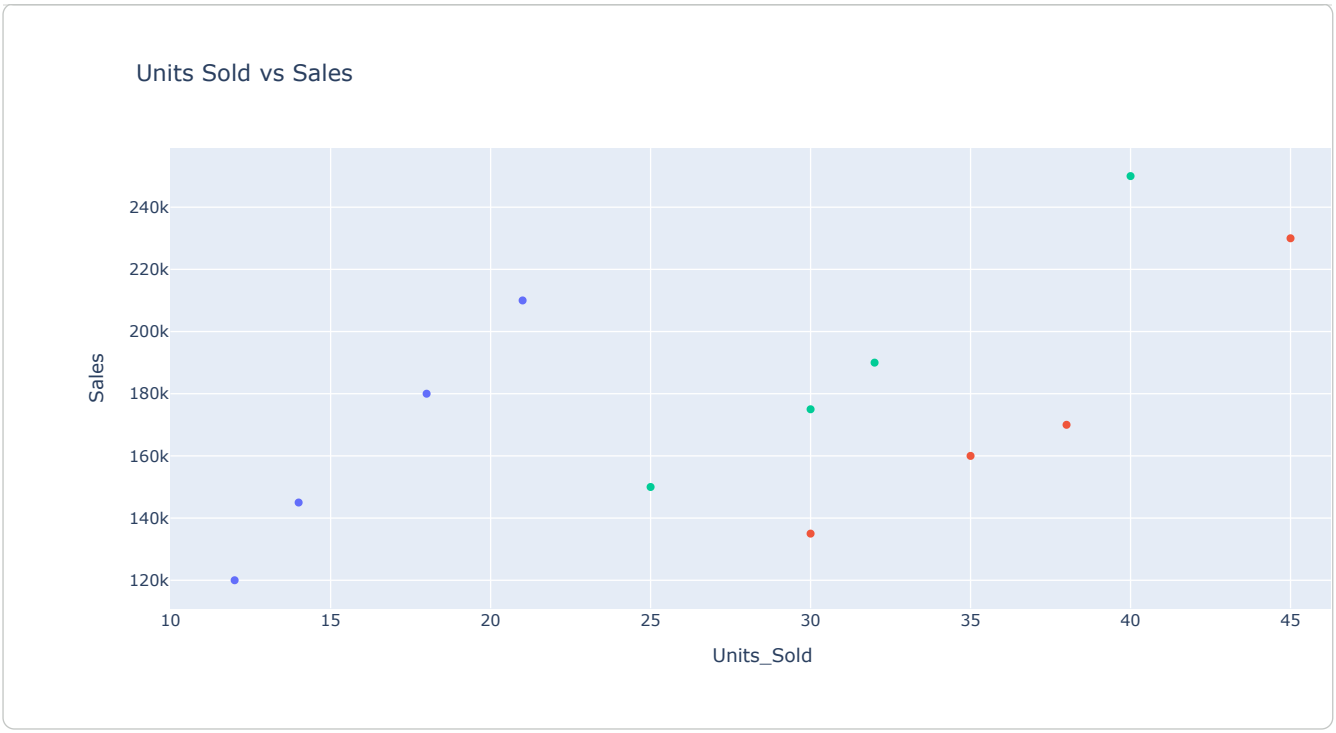
**Step 6 - interactive Scatter plot **

```
# TO analyze the relationship between units sold and sales using an interractive scatter plot.
fig = px.scatter(df, x='Units_Sold', y='Sales', title='Units Sold vs Sales')
fig.show()
```



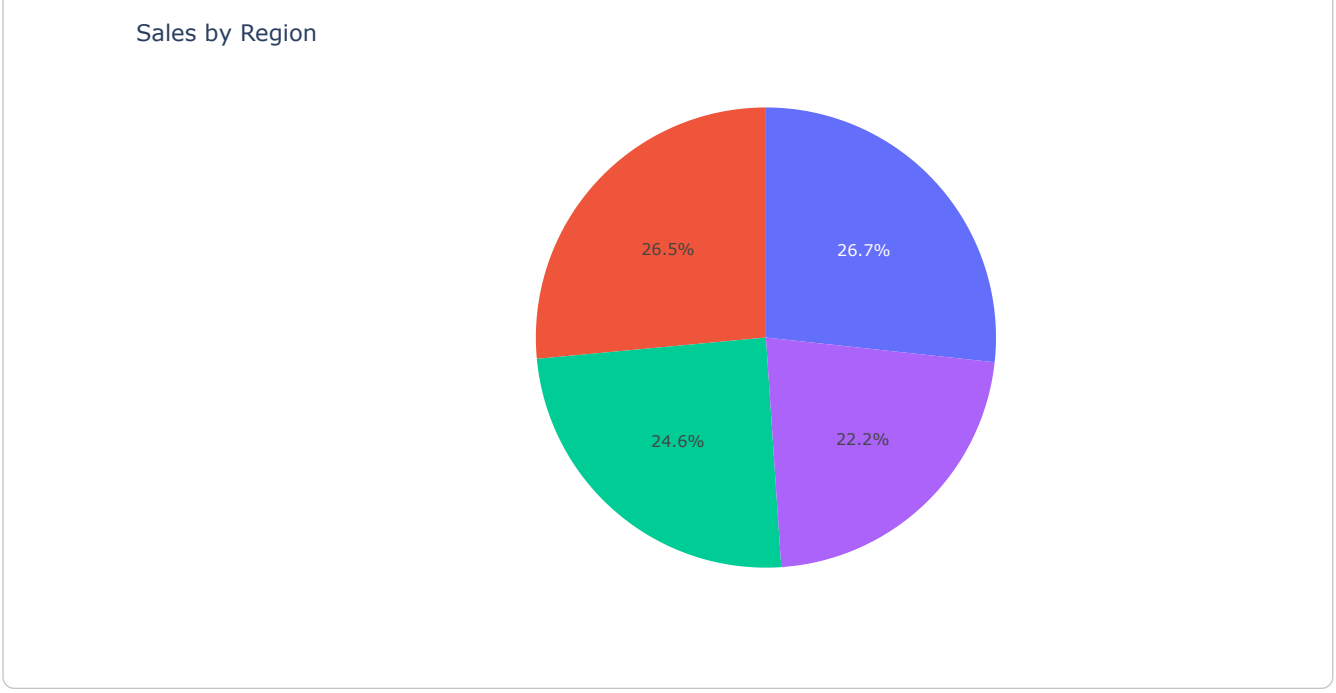
**Step 7 : Interactive Scatter plot with color **

```
# create an scatter plot with color .
fig = px.scatter(df, x='Units_Sold', y='Sales', color='Product',title='Units Sold vs Sales')
fig.show()
```



Step 8 : Create an interactive Pie Chart

```
# To visualize regional sales contribution  using an interactive Pie Chart
fig = px.pie(df, values='Sales', names='Region', title='Sales by Region')
fig.show()
```



Step 9 :Plotly Graph Objects

```
# ploting plotly graph

fig = go.Figure()

fig.add_trace(go.Line(
    x=df['Month'],
    y=df['Sales'],
    mode='lines+markers',
    name='Sales'
))

fig.show()
```

```
/usr/local/lib/python3.13/dist-packages/plotly/graph_objs/_deprecations.py:378: DeprecationWarning:

plotly.graph_objs.Line is deprecated.
Please replace it with one of the following more specific types
  - plotly.graph_objs.scatter.Line
  - plotly.graph_objs.layout.shape.Line
  - etc.
```